MIL-T-46G
29 August 1978
SUPERSEDING
MIL-T-46F
23 July 1971

MILITARY SPECIFICATION

TRAY, MESS, COMPARTMENTED, STAINLESS STEEL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a stainless steel compartmented mess tray (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATION

FEDERAL

PPP-B-636 - Boxes, Shipping, Fiberboard

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Command, ATTN: DRDNA-ES, Natick, MA 01760, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 7350

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads For 40" x 48" Pallets

DRAWING

US ARMY NATICK RESEARCH AND DEVELOPMENT COMMAND

5-13-918 - Tray, Mess, Compartmented, Stainless Steel

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 167 - Stainless and Heat-Resisting Chromium-Nickel Steel Plate, Sheet and Strip

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

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UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), the contractor shall furnish a sample for first article inspection and approval (see 4.2, 6.2, and 6.4).

3.2 Materials. Materials not definitely specified shall be of the quality normally used by the manufacturer in his standard commercial practice provided the completed item complies with all provisions of this specification. Commercial tolerances shall apply to material thicknesses (see 6.3).

3.2.1 Stainless steel sheet or strip. Stainless steel sheet or strip shall conform to type 302 or 304 of ASTM A 167, 24 gauge, 0.0250 inch (0.64 mm).

3.3 Design and construction. The design and construction shall be in accordance with Drawing 5-13-918 and as specified herein. The trays shall be formed seamless from material specified in 3.2.1. The flat, folded bead of the rim shall be closed and mechanically sealed by the crimping or flattening of the rim down to the face of the tray. The trays shall be flat within 3/32 of an inch (2.4 mm) (see 4.5.1 (a)), and all edges shall be straight so as to prevent the insertion of a 1/32 inch (0.79 mm) feeler gauge between all edges and flat surfaces (see 4.5.1). Each tray shall have four circular spacer buttons fabricated of the material specified in 3.2.1 and spot welded one on each corner.

3.4 Finish. The trays shall have a number 2B or better finish. There shall be no discoloration.

3.5 Marking for identification. Trays shall be marked with the manufacturer's name, trade name, or trademark readily identifiable with the manufacturer. Surface opposite markings shall show no penetration.

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3.6 Workmanship. The finished trays shall be clean and shall not contain any pits, scale, rust, blisters, trimming marks, scratches, cracks, fractures, wrinkles or die marks. All sharp edges, burrs, and slivers shall be removed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required, it shall be examined for defects listed in table I, dimensions specified, and tested as specified in 4.5.1. Failure to pass the test shall be cause for rejection of the first article sample. Deficiencies found should be reported to the contractor for correction prior to start of production.

4.4 Quality conformance inspection. Except where otherwise specified herein, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.2 Process examination. Examination shall be made to determine spacer buttons are welded where specified (see 3.3). Whenever nonconformance is noted, correction shall be made to affected items and process.

4.4.3 End item inspection. The lot shall consist of all trays offered for inspection at one time. The sample unit shall be one completely fabricated tray.

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.4.3.1 Visual examination. Examination shall be made of the trays-for defects listed in table I. The inspection level shall be II with an acceptable quality level (AQL) of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

TABLE I. Classification of visual defects

Examine	Defect	Classification Major	Minor
Finish	Not number 2B or better	X	
	Sharp edges, burrs, or slivers	X	
	Not clean, evidence of die marks	X	
Construction, and workmanship (general)	Crack or fracture	X	
	Pits, scale, blisters, wrinkles, or discolorations	X	
	Rust or scratches	X	
	Tray edges not folded and sealed	X	
Spacer button	Missing, edges not rounded	X	
	Not placed where specified (visual determination)	X	
	Not spot welded	X	
Marking for identification	Missing, incomplete, illegible	X	
	Surface opposite marking uneven		X
	Surface opposite marking penetrated	X	

4.4.3.2 Dimensional examination. Examination shall be made of the completed trays to determine compliance with this specification. Any dimension not within the specified requirement shall constitute a defect. The inspection level shall be S-2 with an AQL of 4.0 defects, expressed in terms of defects per hundred units.

4.4.3.3 End item testing. Testing shall be performed on the end item in accordance with 4.5.1. The inspection level shall be S-2, and the AQL shall be 4.0 defects, expressed in terms of defects per hundred units.

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4.4.4 Packaging inspection. An examination shall be made to determine that preservation, packing and marking as required by section 5 are complied with. Defects shall be scored as specified below. The sample unit shall be one shipping container fully packaged, except that it need not be closed. The lot shall consist of the number of containers offered for inspection at one time. The inspection level shall be S-2 and the AQL shall be 2.5 defects, expressed in terms of defects per hundred units.

TABLE II. Examination of packaging

Examine	Defect
Marking (exterior and unit packs)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Material	Component missing, damaged, or otherwise defective.
Workmanship	Bulged or distorted containers.
Contents	Number per container more or less than required.

4.4.4.1 Examination for palletization. An examination shall be made to determine that the palletization complies with the section 5 requirements. Defects shall be in accordance with table III. The sample unit shall be one palletized unit load fully prepared for delivery. The lot shall be the number of palletized unit loads in the end item inspection lot. The inspection level shall be S-1 and the acceptable quality level (AQL) shall be 6.5 defects per hundred units.

TABLE III. Examination of palletization

Examine	Defect
Finished dimension	Length, width or height exceeds specified maximum requirement.
Palletization	Pallet pattern not as specified. Load not bonded with required straps as specified.

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TABLE III. Examination of palletization (cont'd)

Examine	Defect
Weight	Exceeds maximum load limits.
Marking	Omitted; incorrect; incomplete; illegible; of improper size, location, sequence or method of application.

4.5 Tests.

4.5.1 Flatness and straightness. Test for flatness and straightness shall be as follows:

(a) The tray shall be placed face down on a flat surface testing plate to determine compliance with 3.3.

(b) The tray shall be placed face down on a flat surface testing plate with one side of the tray supported by a horizontally placed 1/2-inch (12 mm) diameter rod. An attempt shall be made to insert a 1/32-inch (0.79 mm) thick feeler gauge between the edge of the mess tray and the test plate opposite the side supported by the rod to determine compliance with 3.3. This procedure shall be repeated for the other three sides. Inability of the tray to meet the requirements of 3.3 shall constitute failure of this test.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial, as specified (see 6.2).

5.1.1 Level A. Twelve trays shall be nested and unit packed in a snug-fitting fiberboard box conforming to style FOL (end opening), type CF, variety SW, or type SF, class domestic of PPP-B-636. Each box shall be securely closed in accordance with method III as specified in the appendix of PPP-B-636.

* 5.1.2 Commercial. The trays shall be unit packed to afford adequate protection against damage during shipment from the supply source to the first receiving activity. The unit pack and the quantity per unit pack shall be the same as that used by the contractor for retail distribution.

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5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

* 5.2.1 Level A packing. Thirty-six trays, unit packed as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style FOL (end opening), V2s of PPP-B-636. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced as specified in the appendix of PPP-B-636.

5.2.2 Level B packing. Thirty-six trays, unit packed as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style FOL (end opening), type CF, variety SW, or type SF, class domestic of PPP-B-636. Each shipping container shall be closed in accordance with method II of the appendix of PPP-B-636.

5.2.2.1 Weather-resistant shipping containers. When specified (see 6.2), the shipping container shall be V3c, V3s, or V4s, fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix thereto.

* 5.2.3 Commercial packing. Trays, unit packed as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. The quantity per shipping container shall be the same as that normally used by the contractor for retail distribution. Containers shall be in accordance with Uniform Freight Classification or National Motor Freight Classification, as applicable.

5.3 Palletization. When specified (see 6.2), trays, packed as specified, shall be palletized in accordance with load type I of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L. Pallet patterns shall be in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be affected by reversing the pattern of each course. If the container is of a size which does not conform to any of the pallet patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

5.4 Marking. In addition to any special marking required by the contract, unit packs and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The trays covered by this specification are intended for use in mess halls of the Armed Forces.

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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) When a first article is required (see 3.1, 4.2, and 6.4).
(c) Selection of applicable levels of preservation and packing (see 5.1 and 5.2).
(d) When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.1).
(e) When palletization is required (see 5.3).

* 6.3 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.2).

6.4 First article. When a first article is required, it should be inspected and approved under the appropriate provisions of ASPR 7-104.55. The first article should be a preproduction sample and should consist of one unit. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for inspection and approval of the first article.

6.5 Metric equivalents. Metric equivalents, indicated in parentheses throughout this document, are based on practices, conversion factors, and symbols specified in ASTM E 380 Standard for Metric Practice, and are for information only. In each instance, the value stated in US customary units shall be controlling.

6.6 Changes from previous issue. The margins of this specification are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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DOCUMENT IDENTIFIER AND TITLE
MIL-T-46G TRAY, MESS, COMPARTMENTED, STAINLESS STEEL

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING

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